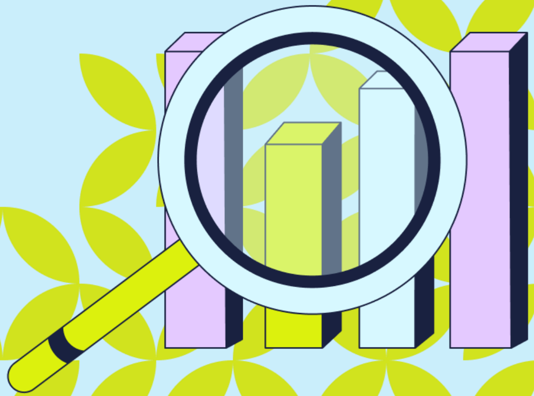
Clockster Documentation

Exposure-aware attendance scoring with SQL & Power BI

Roy Padayao



Project Overview

- This project analyzes employee attendance data from HR department to identify employee discipline, violation trends, and department-level employee management.
- The analysis required combining data from multiple sources and producing a shift-based discipline scoring model using SQL and Power BI.

Data source:

Name	Date modified	Type	Size
attendance	08/11/2022 4:23 pm	Microsoft Excel Co...	695 KB
leave_requests	08/11/2022 4:23 pm	Microsoft Excel Co...	5 KB
payroll	08/11/2022 4:23 pm	Microsoft Excel Co...	34 KB
schedules	20/01/2025 7:16 pm	Microsoft Excel Co...	627 KB
users	08/11/2022 4:23 pm	Microsoft Excel Co...	7 KB

Excel

Raw data initial exploration.

- Inspection of column types, date formats, duplication, and missing values.
- Exploration only; final cleaning moved to SQL for reproducibility.

Leave Requests

non
atomic
values

user_id	first_name	last_name	type	leave_type	dates	time_start	time_end	timezone	status	created_at
			leave	day_off	["2021-10-27"]				accepted	25/10/2021 12:38
			leave	day_off	["2021-11-12"]				accepted	07/11/2021 15:20
74465			leave	unpaid	["2021-11-16"]				accepted	15/11/2021 13:45
75839			leave	day_off	["2021-11-24"]				accepted	16/11/2021 2:18
74042			leave	day_off	["2021-11-28"]				accepted	27/11/2021 7:06
74054			leave	sick	["2021-12-23"]				accepted	23/12/2021 4:04
74049			leave	sick	["2021-12-09", "2021-12-10", "2021-12-11", "2021-12-12", "2021-12-13", "2021-12-14"]				accepted	27/12/2021 13:50
74044			leave	sick	["2022-01-07", "2022-01-08"]				accepted	07/01/2022 2:25
74042			leave	annual	["2022-03-11"]				accepted	01/03/2022 7:45
74461			leave	compensatory	["2022-06-30"]				accepted	28/05/2022 7:08
83885			leave	annual	["2022-06-03"]				accepted	28/05/2022 7:49
88348			leave	day_off	["2022-06-07", "2022-06-08", "2022-06-09"]				accepted	03/06/2022 10:02
83902			leave	day_off	["2022-06-07"]				accepted	03/06/2022 12:25
83902			leave	day_off	["2022-06-17"]				accepted	09/06/2022 4:29
88348			leave	day_off	["2022-06-14"]				accepted	13/06/2022 9:22
74042			leave	compensatory	["2022-06-28", "2022-06-29"]				accepted	16/06/2022 8:33
88357			leave	sick	["2022-06-23", "2022-06-24"]				accepted	23/06/2022 13:04
74049			leave	sick	["2022-06-20", "2022-06-21", "2022-06-22", "2022-06-23", "2022-06-24", "2022-06-25", "2022-06-26", "2022-06-27", "2022-06-2				accepted	27/06/2022 23:14
74042			leave	compensatory	["2022-06-30"]				accepted	29/06/2022 10:27
88357			leave	day_off	["2022-07-11"]				accepted	11/07/2022 0:47
88357			leave	compensatory	["2022-07-12", "2022-07-13"]				accepted	12/07/2022 13:11
88357			leave	unpaid	["2022-07-14"]				accepted	14/07/2022 1:47
74461			leave	compensatory	["2022-07-27"]				accepted	18/07/2022 8:28
88357			leave	sick	["2022-07-15", "2022-07-16", "2022-07-17", "2022-07-18", "2022-07-19", "2022-07-20"]				rejected	19/07/2022 11:20
74042			leave	compensatory	["2022-08-01", "2022-08-02"]				accepted	22/07/2022 8:44
83902			leave	day_off	["2022-07-29"]				accepted	26/07/2022 4:16
74461			leave	compensatory	["2022-09-27", "2022-09-28", "2022-09-29", "2022-09-30"]				accepted	30/07/2022 3:35
79765			leave	annual	["2022-08-07"]				accepted	01/08/2022 9:04
74052			leave	compensatory	["2022-08-06"]				accepted	01/08/2022 9:43
74049			leave	compensatory	["2022-08-13"]				accepted	07/08/2022 23:54
88348			leave	day_off	["2022-08-10"]				accepted	09/08/2022 2:31
74042			leave	compensatory	["2022-08-20"]				accepted	15/08/2022 1:01
120666			leave	compensatory	["2022-08-21"]				accepted	16/08/2022 8:18
83885			leave	compensatory	["2022-08-27"]				accepted	18/08/2022 8:57

Attendance

user_id	first_name	last_name	location	date	time	timezone	case	source
74025				20/10/2021	20:08:45	+08:00	IN	mobile
74025				20/10/2021	20:09:11	+08:00	IN	mobile
74027				20/10/2021	22:29:56	+08:00	IN	mobile
74027				21/10/2021	0:15:33	+08:00	OUT	mobile
74025			Nu Orange	21/10/2021	13:22:20	+08:00	IN	mobile
74027			Nu Orange	21/10/2021	15:47:52	+08:00	IN	mobile
74135			Nu Orange	21/10/2021	18:09:37	+08:00	IN	mobile
74025			Nu Orange	22/10/2021	13:40:57	+08:00	IN	mobile
74138				22/10/2021	14:33:25	+08:00	IN	mobile
74027			Nu Orange	22/10/2021	16:15:48	+08:00	IN	mobile
74050			Clinic	25/10/2021	13:51:35	+08:00	IN	mobile
74027				25/10/2021	13:58:21	+08:00	IN	mobile
74053			Clinic	25/10/2021	14:20:04	+08:00	IN	mobile
74049			Clinic	25/10/2021	14:20:16	+08:00	IN	mobile
74050				25/10/2021	14:20:21	+08:00	IN	mobile
74465			Nu Orange	25/10/2021	14:20:25	+08:00	IN	mobile
74054			Nu Orange	25/10/2021	14:20:29	+08:00	IN	mobile
74556			Nu Orange	25/10/2021	14:20:29	+08:00	IN	mobile
74053			Clinic	25/10/2021	14:20:44	+08:00	IN	mobile

Schedules

non
atomic
values

sched_typ	dates	time_start	time_end	timezone	time_plan	break_tim	leave_type	user_id
free	["2021-10-	9:00:00	8:59:00	+08:00	7200	0		{74138,74025,74027,74062,74135}
free	["2021-10-	9:00:00	8:59:00	+08:00	7200	0		{74138,74025,74027,74062,74135}
free	["2021-10-	9:00:00	8:59:00	+08:00	7200	0		{74138,74025,74027,74062,74135}
free	["2021-10-	9:00:00	8:59:00	+08:00	7200	0		{74138,74025,74027,74062,74135}
free	["2021-10-	9:00:00	8:59:00	+08:00	7200	0		{74138,74025,74027,74062,74135}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	9:00:00	18:00:00	+08:00	28800	3600		{125721,125744,125748,126082,129675,132484,138225,146166,146182,155509,157837,157916,159207,159217,160306}
work	["2021-10-	15:00:00	22:00:00	+08:00	21600	3600		{74042}
work	["2021-10-	9:00:00	20:00:00	+08:00	36000	3600		{74052}
work	["2021-09-	15:00:00	22:00:00	+08:00	25200	0		{74461}
work	["2021-10-	8:30:00	14:30:00	+08:00	21600	0		{74044}
work	["2021-10-	15:30:00	21:30:00	+08:00	21600	0		{74044}
work	["2021-10-	15:00:00	22:00:00	+08:00	25200	0		{74050}
work	["2021-10-	8:00:00	15:30:00	+08:00	27000	0		{74050}
work	["2021-10-	8:00:00	15:00:00	+08:00	21600	3600		{74052}

SQL

Data cleaning using SQL

- Unnest and Splitting columns to rows

Query | Query History

```sql
WITH leave_req1_w_row_num AS (
    SELECT DISTINCT
)
```

Data Output | Messages | Notifications

	user_id text	first_name text	last_name text	type text	leave_type text	dates text	time_start text	time_end text	timezone text	status text	created_at text
41	74049			leave	compensatory	[2022-09-13]	[null]	[null]	[null]	accepted	2022-09-0
42	88357			leave	unpaid	[2022-09-08, 2022-09-09, 2022-09-10]	[null]	[null]	[null]	accepted	2022-09-0
43	79765			leave	annual	[2022-09-20]	[null]	[null]	[null]	accepted	2022-09-1
44	79738			leave	compensatory	[2022-09-28]	[null]	[null]	[null]	accepted	2022-09-2
45	74042			leave	compensatory	[2022-09-28]	[null]	[null]	[null]	accepted	2022-09-2
46	74042			leave	compensatory	[2022-10-11, 2022-10-12]	[null]	[null]	[null]	accepted	2022-10-0
47	88348			leave	annual	[2022-10-10]	[null]	[null]	[null]	accepted	2022-10-0
48	74052			leave	compensatory	[2022-10-09]	[null]	[null]	[null]	accepted	2022-10-0
49	74049			leave	compensatory	[2022-10-14]	[null]	[null]	[null]	accepted	2022-10-0
50	83884			leave	compensatory	[2022-10-22, 2022-10-23]	[null]	[null]	[null]	accepted	2022-10-1
51	74049			leave	compensatory	[2022-10-21]	[null]	[null]	[null]	pending	2022-10-1

Total rows: 51 of 51 | Query complete 00:00:00.062 | Ln 2, Col 17

Query | Query History

```sql
WITH leave_req1_w_row_num AS (
    SELECT DISTINCT *,
        ROW_NUMBER() OVER() AS row
    FROM leave_req1
),
dates_table AS (
    SELECT row, UNNEST(string_to_array(dates, ',')) AS date
    FROM leave_req1_w_row_num
)
SELECT
    l.user_id,
    l.type,
    l.leave_type,
    l.status,
    l.created_at,
    to_date(REPLACE(REPLACE(dt.date, '[', ''), ']', ''), 'YYYY-MM-DD') AS date
FROM leave_req1_w_row_num l
LEFT JOIN dates_table dt ON dt.row = l.row;
```

Data Output | Messages | Notifications

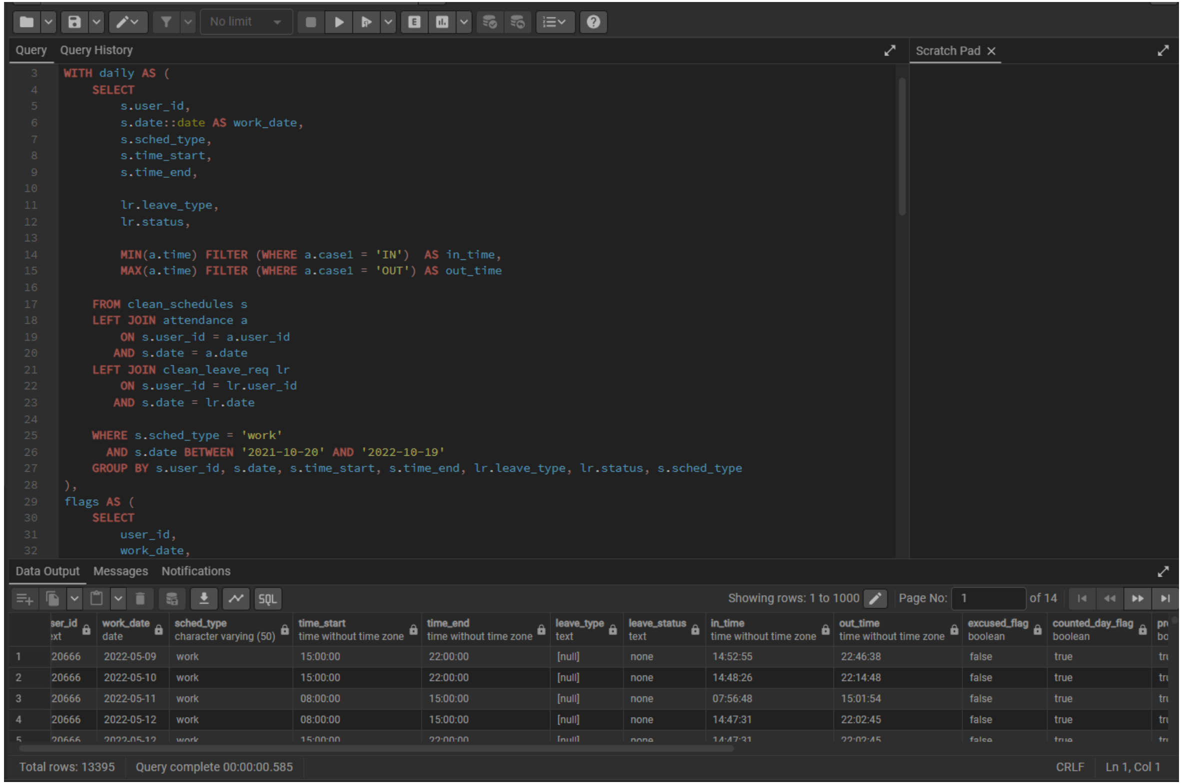
	user_id text	type text	leave_type text	status text	created_at text	date date
1	74044	leave	sick	accepted	2022-01-07 02:25:11	2022-01-07
2	74044	leave	sick	accepted	2022-01-07 02:25:11	2022-01-08
3	83885	leave	compensatory	accepted	2022-08-18 08:57:08	2022-08-27
4	83902	leave	day_off	accepted	2022-07-26 04:16:11	2022-07-29
5	88357	leave	sick	rejected	2022-07-19 11:20:32	2022-07-15

Total rows: 92 of 92 | Query complete 00:00:00.060

Normalization

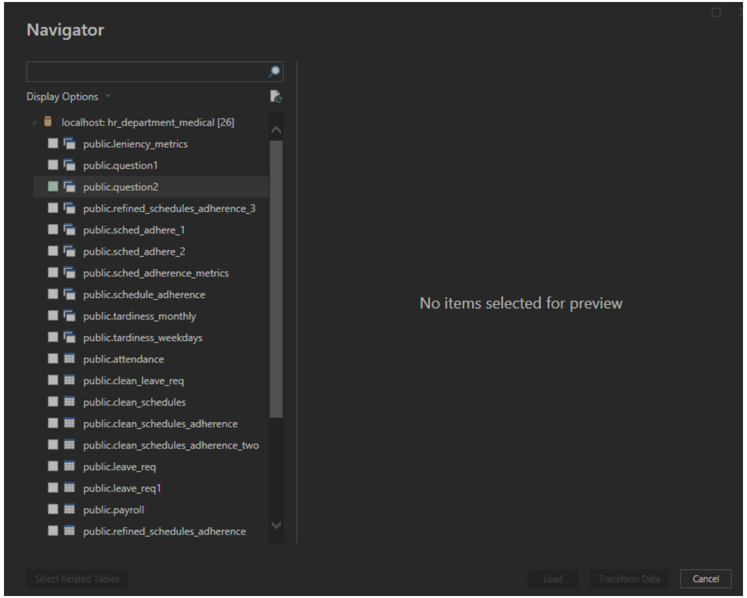
- Collapse duplicate logs per day
- Joining tables
- Creating Shift-Level Flags

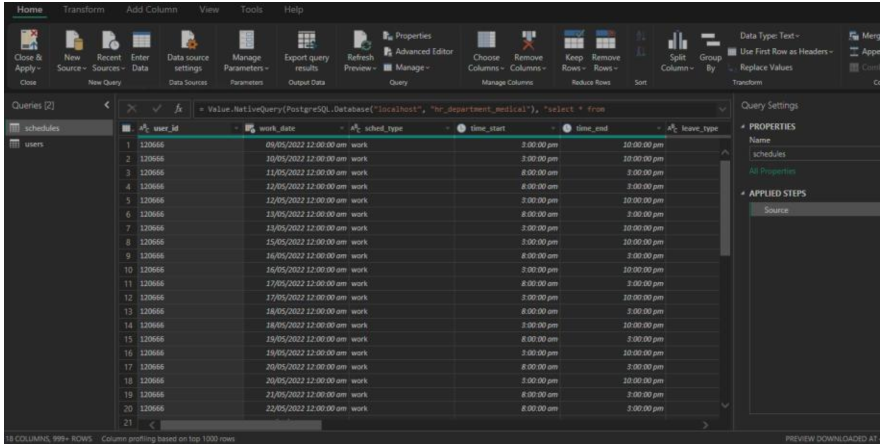
Flag	Meaning
counted_day_flag	Scheduled day AND not accepted/pending leave
excused_flag	Accepted/Pending leave
late_in_flag	Actual in > 10 mins AND < 2 hours
early_out_flag	Actual out < −10 mins AND > −2 hours
unexcused_absence_flag	No in/out AND not excused
incomplete_punch_flag	Only one of IN/OUT present

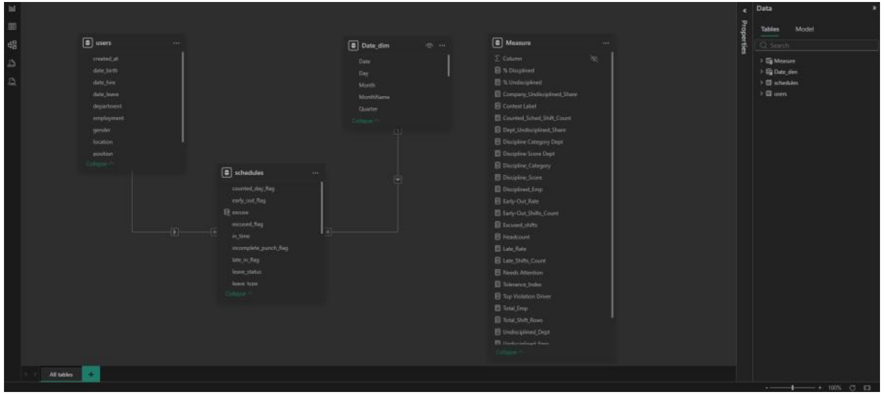


Data Modeling

- Importing the transformed and shaped data tables from SQL to Power BI



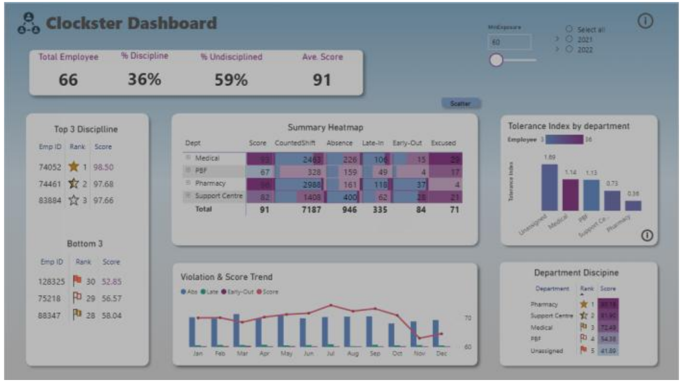




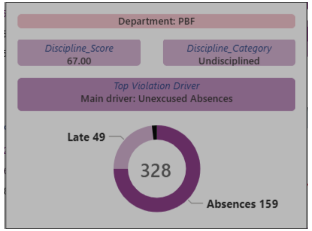
Visualization

- Creating dashboard to visualize and analyze data.

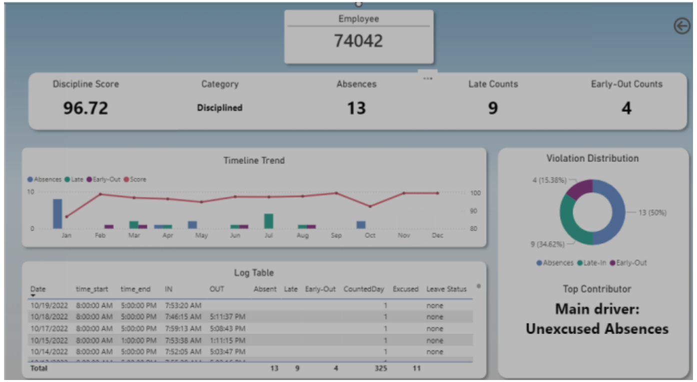
Executive Summary



Tooltip



Employee Drillthrough page



Department Drillthrough page